COMPUTATIONAL ORDINAL SOCIOMETRY (NEXORD): Scientific Manifesto, Theoretical-Methodological Foundations, and Open Science Governance

Towards a Physics of the Group: Mathematical Resolution of the Dyadic Vote, Spectral Topology, and Socio-Thermodynamic Diagnosis

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01. SCIENTIFIC ABSTRACT

The NEXORD Scientific Manifesto formalizes the paradigmatic transition from classical Morenian sociometry (founded in 1934 on binary nominal counts and unweighted scales) toward an **authentic mathematical physics and social thermodynamics of human systems**. Facing information collapse, non-decision artifacts, and the analytical distortion of undifferentiated voting, NEXORD introduces a rigorous axiomatic framework based on the natural order of preferences (ordinal matrices *SMIb*), full dyadic projection in Hilbert spaces and Grassmannian manifolds Gr(k, N), and the exact deduction of factorial duality between emission and reception.

02. THE FOUR FOUNDATIONAL INNOVATIONS

1. Four-Position SMIb Tetragram

Transcends univariate scalar matrices through an inalienable 4-dimensional dyadic cell: **A1 (p_DA)**: sender's expectation of partner's action; **A2 (DA)**: effective preference granted; **A3 (RECEIVES)**: effective preference received from partner; and **A4 (p_RECEIVES)**: partner's expectation of sender's action.

2. Exponential Physical Law W(k) = 2^(1-k)

Continuous gravitational decay: 1st=1.000, 2nd=0.500, 3rd=0.250, 4th=0.125... Universal grid **Q81 (9×9)** systematically encoding all 81 possible dyadic interaction states without blind spots or forced neutrality.

3. Factorial Duality & Barycentric Theorem

Rigorous decoupling: **Correspondence Analysis (CA)** on summed discrete frequencies (**F_ACT = F_DA + F_RC**) via χ^2 metrics; and **Principal Component Analysis (PCA)** on the continuous **SDR** balance matrix via SVD. Proof of the **Barycentric Theorem (Δ = 0.000)**.

4. Spectral Topology in Grassmann Gr(k,N)

Relational X-ray diagnostics: Estrada operators (*EE*), Fiedler algebraic connectivity (λ_2), Heider structural frustration (λ_1), and Tononi dyadic integrated information Φ to quantify gravitational strain and systemic health.

💡 **Strict Barycentric Theorem (Δ = 0.000):** In any factorial space, the subject's active coordinates lie with exact analytical precision at the weighted center of mass (barycenter) between their emission projection (DA / SDA) and reception projection (RC / SRC). This provides the physical grounding for the dyadic biplot: the relational arrow describes the true dialectical vector between expansiveness and sociometric status.

03. STRUCTURAL MAP OF THE CENTRAL MANIFESTO (12 SECTIONS)

The main body of the Manifesto (Sections 1 to 7) establishes the epistemological and mathematical grounding of the framework:

- **§1. Introductory Framework and Breakdown of the Classical Paradigm:** Moreno and Lewin's replicability crisis and the requirement of ordinal metrics.
- **§2. Axiomatics of the SMIb Tetragram and Q81 Grid:** Formal definition of the 4 positions and discrete dyadic state algebra.
- **§3. Physical Modulators and Social Thermodynamics:** Shannon entropy, relational viscosity, absolute (BDR) and relative density (SDR).
- **§4. Canonical Factorial Duality:** Methodological separation between Benzécri χ^2 (CA) and continuous SVD (PCA), dyadic barycenters, and biplots.
- **§5. Standardized Battery VAR10 and Adjectival Attribution AAG16:** Canonical variables and 4 factors (THEME, FUNC, COGN, AFFEC) on forced even scales (1 to 8).
- **§6. Algorithmic Pipeline & Factual Execution:** Deterministic in silico workflow, Monte Carlo ingestion, and empirical consistency validation.
- **§7. Ethical Governance, Open Source, and FAIR Philosophy:** Data deontology, self-esteem protection, and UNESCO transfer model.

04. SPECIFICATION OF THE 5 FORMALIZED TECHNICAL ANNEXES (SECTION 8)

Technical Annex	Formal Domain	Methodological Contribution & Demonstrated Invariant
Annex I	Moreno ↔ Breve Transposition	Exact equivalence matrix M_ORD. Bijective algorithm converting ordinal rankings into Morenian frequencies without spurious heuristics.
Annex II	6D / 4D Factorial Projection	Canonical factorial axes, DIMO axonometric projections, and dyadic barycenter decomposition in affine spaces.
Annex III	18 Relational Densities	The 3 canonical prisms (Affinity, Strain, and Metaperception) derived from gravitational law W(k) = 2^(1-k) across emission and reception.
Annex IV	Wavelet FFT Spectrogram	Fast Fourier Transform applied to relational order. Demultiplexing status carrier waves from high-frequency friction harmonics.
Annex V	Markov Chains & Status	Stochastic transition matrices, Perron-Frobenius principal eigenvector derivation, and asymptotic role convergence.

05. SUPPLEMENTARY DOCUMENTATION & CONCEPTUAL GLOSSARY

- **20-Plate High-Res Catalog (Part III)**
  Full graphic and tabular atlas: from the Moreno-Lewin historical diptych to 18D spectrograms and multi-lane factorial biplots.
- **General Glossary (Part IV)**
  Comprehensive lexicon featuring 42 formalized clinical sociometric operators, standardized acronym tables, and explicit formulas.
- **Responses to Critiques (Part V)**
  Systematic counter-arguments to 10 standard methodological objections (physicalism, determinism, ethics) and AI dialogue transcripts.
- **Thematic Bibliography (Part VI)**
  Structured academic corpus across 6 axes: foundational sociometry, multivariate analysis, graph theory, and complex social network physics.

06. ETHICAL GOVERNANCE & INVIOLABLE DEONTOLOGICAL COMMITMENT

When applied to human collectives in real-world environments (educational, organizational, clinical, or community settings), NEXORD enforces **Privacy by Design** and a strict **formative, non-punitive mandate**:

- **Mandatory Canonical Anonymization:** All participants are systematically anonymized via dissociated matrix tokens (1A..5A for Group A, 1B..5B for Group B), preserving absolute mathematical voting secrecy.
- **Strict Prohibition of Self-Esteem Harm:** It is deontologically forbidden to attribute to individual personality traits what social thermodynamics demonstrates to be systemic group equilibria or tension states. All diagnostic outputs must facilitate group health, never stigmatize individuals.
- **CC-BY-4.0 Open License and Determinism:** Full scientific transparency for international research and teaching, banning non-reproducible black-box algorithms.

07. FILE INVENTORY OF THE CERN ZENODO MASTER RECORD

The permanent unified record (**DOI: 10.5281/zenodo.23329642**) organizes the following downloadable and interconnected assets:

Filename	Format / Size	Scientific Function in the Ecosystem
00_RESUMEN_EJECUTIVO_MANIFIESTO_NEXORD.pdf	PDF (~150 kB)	**Default Preview File.** Executive synthesis, abstract, and innovation roadmap (this document).
MANIFIESTO_Y_ANEXOS_NEXORD_2026_ES_EN.html	HTML (~410 kB)	**Master Bilingual Interactive Document.** Full manifesto (12 §§), 5 annexes, catalog, dynamic glossary, and critiques.
NEXORD_ANEXO_MATEMATICO_v10.pdf / _EN.pdf	PDF (~35 kB)	Formal specification of relational densities, SDR/BDR invariants, and mathematical proofs.
NEXORD_RESPUESTAS_CRITICAS_v91_ES.pdf / _EN.pdf	PDF (~265 kB)	Responses to 10 epistemological critiques and transcripts of sustained methodological AI debates.
NEXORD_GLOSARIO_v911_ES.pdf / _EN.pdf	PDF (~185 kB)	Conceptual lexicon of 42 clinical sociometric concepts and canonical operator reference tables.

08. INTERACTIVE VISORD LABORATORIES (LEVEL 8)

The ecosystem integrates four virtual laboratory exploration suites accessible directly from the institutional web platform:

- **VISORD_demo**
  Continuous 3D/4D group dynamics simulator, phase transitions, and social thermodynamics for N < 51.
- **VISORD_cultural**
  Dramatic sociometry applied to master narratives (*12 Angry Men*, *Bernarda Alba*, *Lord of the Flies*).
- **VISORD_radiological**
  Relational X-rays (15-27-54), Grassmannian manifold topology, and invisible ostracism detection.
- **VISORD_dialectical**
  Real-time micro-interactions, common-pool resource dilemmas, and Bales IPA ↔ SMIb Moviola engine.